

Exploratory Data Analysis on the Titanic Dataset

Objective

To explore the Titanic dataset and understand the patterns behind passenger survival using visual and statistical methods.

Dataset Overview

Source: `train.csv`

Key columns: `Pclass`, `Name`, `Sex`, `Age`, `SibSp`, `Parch`, `Ticket`, `Fare`, `Cabin`, `Embarked`

Visual Insights

1. Survival Count

- More passengers did **not** survive (around 60%).
- Plot: Bar chart of survival (0 vs. 1)

2. Gender vs. Survival

- Significantly more **females survived** compared to males.
- Plot: Countplot of `Sex` vs `Survived`

3. Pclass and Gender vs. Survival

- Higher survival in **1st class**, especially for females.
- Lowest survival in 3rd class males.
- Plot: Countplot of `Pclass` with `hue='Sex'`

4. Age Distribution

- Wide range of passengers; children and young adults had slightly higher survival.
- Plot: Histogram / KDE of `Age` with survival hue

5. Fare vs. Class/Survival

- Passengers who paid **higher fares** were more likely in **1st class** and had **higher survival**.
- Plot: Boxplot of `Fare` grouped by `Pclass` and `Survived`

6. Correlation Matrix

- Strong negative correlation between `Sex` (encoded) and `Survived`
- `Fare` has mild positive correlation with survival.

👁 Visual Plot Summary.

Gender vs. Survival(countplot)

Observations:

- A much higher proportion of females survived compared to males.
- Majority of males did not survive.
- This indicates gender was a significant factor in survival – possibly due to the "women and children first" evacuation rule during the Titanic disaster.

Pclass & Sex vs. Survival (Grouped Countplot)

Observations:

- 1st class passengers had the highest survival, especially females.
- 3rd class males had the lowest survival.
- Suggests both class and gender together played a major role in survival.

Fare Distribution Histogram

Observations:

- Fare distribution is right-skewed (most passengers paid lower fares).
- Few outliers with very high fare values (indicating 1st class passengers).
- Could consider log transformation of Fare to normalize distribution before modeling.

Age Distribution Histogram

Observations:

- Age ranges from infants to seniors.
- Distribution is slightly right-skewed.
- A large number of passengers were in the 20–30 age group.
- Missing ages (from earlier EDA) can be imputed with median or predicted from similar groups (e.g., title, Pclass).

Age vs. Survival Scatter Plot

Observations:

- Younger passengers (especially children) have a higher chance of survival.
- Many adult males in their 20s and 30s did not survive.
- The scatter gives a good sense of how survival varies across age groups

Fare vs. Survival Scatter Plot

Observations:

- Passengers with higher fares generally had better survival rates.
- This again reflects the privilege of upper-class (1st class) travelers.
- Lower fares are mostly associated with non-survivors.

Correlation Heatmap

Observations:

- Sex has a strong negative correlation with Survived → meaning females (encoded as 0 or 1) were more likely to survive.
- Fare has a positive correlation with Survived.
- Pclass has a negative correlation with survival, confirming that higher classes (lower Pclass value) had better chances.